22CH203 – ENGINEERING CHEMISTRY II

UNIT V: NUCLEAR REACTIONS

Lesson Plan 3: Nuclear reactions: Fission and fusion - Radiocarbon dating

TOPIC: Nuclear reactions- introduction, types with examples

1. NUCLEAR REACTIONS- INRODUCTION

A nuclear reaction refers to a transformation of a target atomic nucleus by bombarding it with projectiles of light nuclei or free nucleons of adequate energy.

Nuclear reactions involve processes where the nucleus of an atom undergoes a change, leading to the formation of different elements or isotopes. These reactions can release enormous amounts of energy and are the driving force behind nuclear power plants, nuclear weapons, and many natural phenomena.

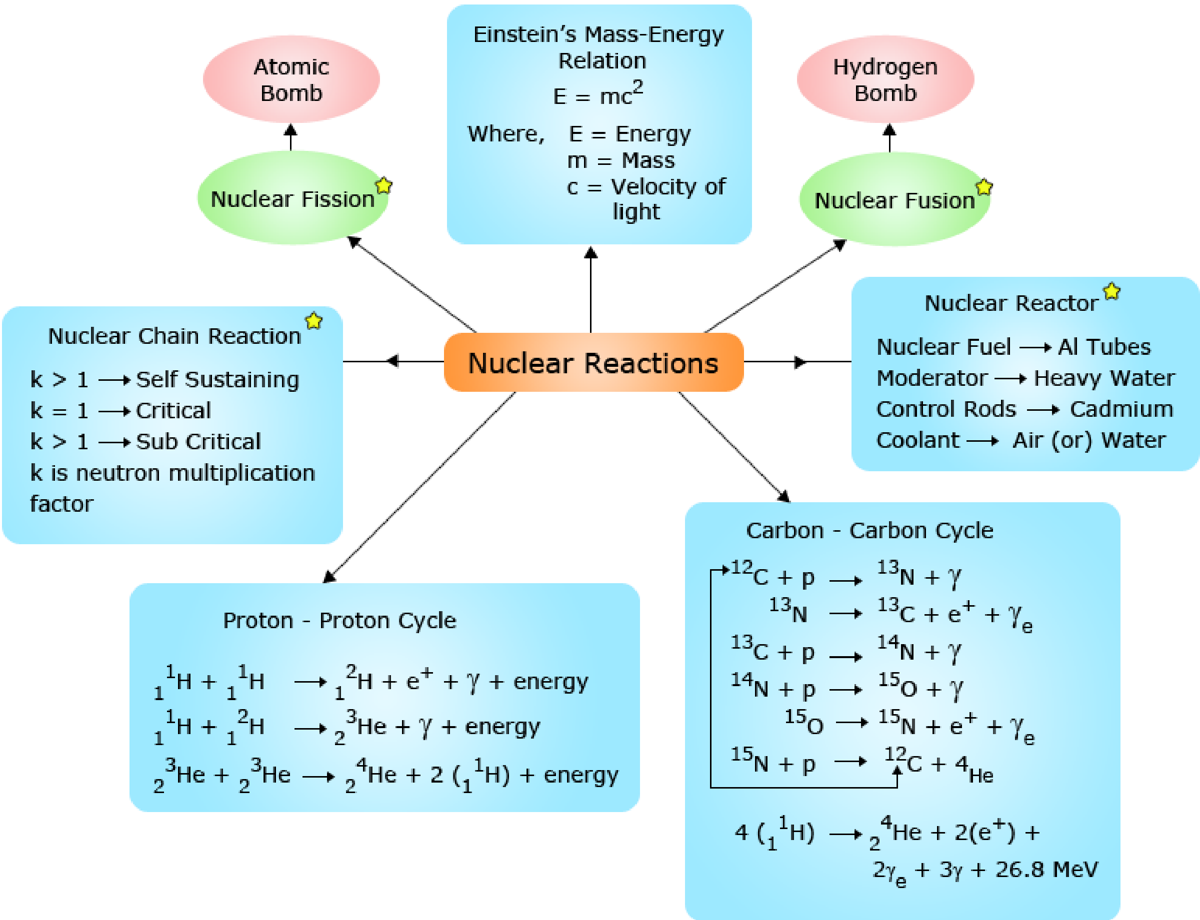


Figure. Nuclear reactions

Types of Nuclear Reactions: The nuclear reactions are classified into the following groups in terms of the overall energy transformation that occurs.

a) Projectile capture reactions:

The bombarding particle is absorbed with or without the emission of gamma radiation. Neutron capture is common but proton capture is also observed in the case of target nuclei with low mass numbers.

$$^{85}_{37}Rb + ^{1}_{0}n \longrightarrow ^{86}_{37}Rb + \gamma$$

$$^{12}_{6}C + ^{1}_{1}H \longrightarrow ^{13}_{7}N + \gamma$$

b) Projectile capture particle emission reactions

In this case, in addition to the product nucleus a massive particle, viz. a-particle, proton and neutron is also produced. The nature of the emitted particle is largely dependent upon the energy of the projectile. Typical examples are given below.

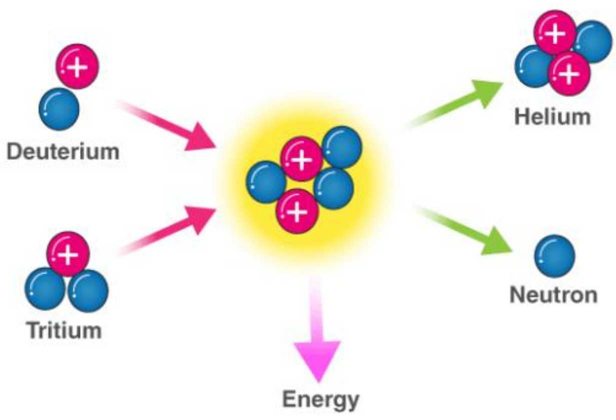
$$^{14}_{7}N + ^{1}_{0}n \longrightarrow ^{14}_{6}C + ^{1}_{1}H$$

$$^{19}_{9}F + ^{1}_{0}n \longrightarrow ^{16}_{7}N + ^{4}_{2}He$$

$$^{11}_{5}B + ^{1}_{1}H \longrightarrow ^{11}_{6}C + ^{1}_{0}n$$

2. FUSION REACTIONS

In a fusion reaction two light weight nuclei combine or fuse to form a single heavy nucleus. For example, two nuclei of deuterium undergo nuclear fusion to yield a heavy nucleus of helium. The fusion reaction takes place at a temperature of about 100 million°C.



$$2\,{}_{1}^{2}\mathrm{H} \longrightarrow {}_{2}^{3}\mathrm{He} + {}_{0}^{1}\mathrm{n} + \text{energy}$$

Fusion reactions are also called thermonuclear reactions since they occur at extremely high temperatures. In a fusion reaction, the mass of the reacting nuclei is greater than that of the nucleus formed:

$$\underset{3.01495\ \text{amu}}{{}_{1}^{3}\mathrm{H}} + \underset{1.00782\ \text{amu}}{{}_{1}^{1}\mathrm{H}} \longrightarrow \underset{4.00260\ \text{amu}}{{}_{2}^{4}\mathrm{He}} + \text{energy}$$

The total mass of the reactants is 4.02277 amu which is 0.02017 amu greater than the mass of the product. The loss in mass results in the release of large amount of energy. The hydrogen bomb is a device which makes use of the principles of nuclear fusion.

3. FISSION REACTIONS

The splitting of a heavy nucleus into two or more smaller nuclei is called nuclear fission. The smaller nuclei formed as a result of fission are called fission products. The process of fission is always accompanied by the ejection of two or more neutrons with the liberation of a large amount of energy.

In 1939, Hahn and Strassmann discovered that a heavy atomic nucleus such as uranium (^{235}U) upon bombardment by a neutron splits into two or more nuclei. ^{235}U first absorbs a neutron to form an unstable compound nucleus. The unstable compound nucleus then divides into two daughter nuclei with the release of neutrons and large amount of energy. Hence nuclear fission is defined as the splitting of a heavy nucleus into two or more smaller nuclei:

$${}_{92}^{235}\mathrm{U} + {}_{0}^{1}\mathrm{n} \longrightarrow {}_{92}^{236}\mathrm{U} \longrightarrow {}_{56}^{139}\mathrm{Ba} + {}_{36}^{94}\mathrm{Kr} + 3\,{}_{0}^{1}\mathrm{n} + \text{energy}$$

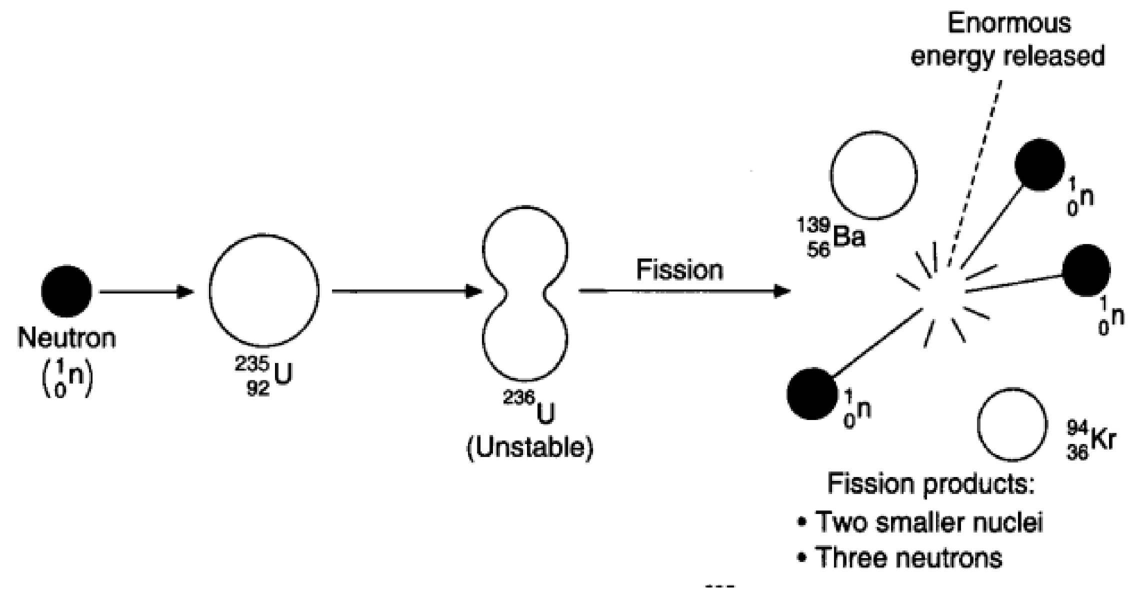


Fig. Fission of ^{235}U.

A given large nucleus can fission in many ways forming a variety of products. For example, the fission of uranium occurs in about 35 ways. Three of them are given as follows:

$$^{235}_{92}U + ^1_0n \rightarrow ^{139}_{56}Ba + ^{94}_{36}Kr + 3\,^1_0n + \text{energy}$$

$$^{235}_{92}U + ^1_0n \rightarrow ^{144}_{54}Xe + ^{90}_{38}Sr + 2\,^1_0n + \text{energy}$$

$$^{235}_{92}U + ^1_0n \rightarrow ^{106}_{42}Mo + ^{128}_{50}Sn + 2\,^1_0n + \text{energy}$$

During this reaction three neutrons are produced. These three neutrons in tum may cause fission in three uranium nuclei producing nine neutrons. These nine neutrons in turn may cause fission in nine uranium nuclei producing 27 neutrons and so on. This process of propagation of the reaction by multiplication in threes at each fission is referred to as chain reaction (Figure 7). A chain reaction will continue until the splitting of all the ^{235}U nucleus. It is observed that not all the neutrons released in the reactions are used up in propagating the chain reaction. Some neutrons may miss their targets and some escape into air and are thus lost. Thus for a chain reaction to occur, the size of the fissionable material should be large enough to capture the neutron. If the size is too small, more neutrons will escape from its surface thereby breaking the chain. The minimum mass of fissionable material required to sustain a chain reaction is called critical mass. The critical mass varies for each reaction. For ^{235}U fission reaction, the critical mass is about 10 kg.

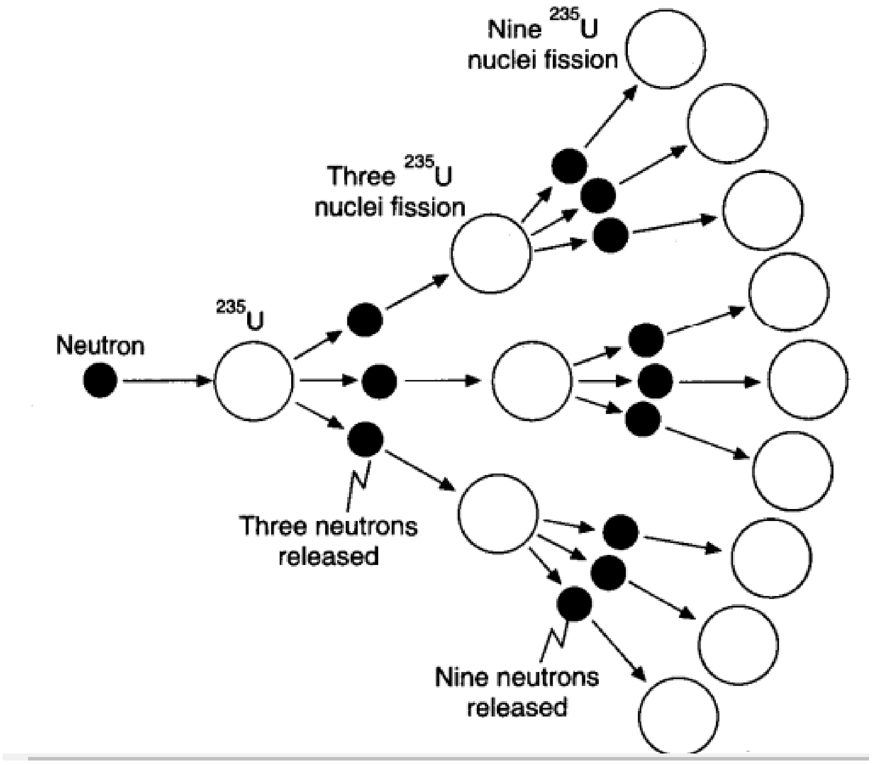


Fig. Propagation of ^{235}U fission reaction.

This chain reaction can be controlled by absorbing a desired number of neutrons so that on an average only one neutron remains available for carrying out further fission. Such a reaction is called controlled chain reaction and the energy released in such a controlled chain reaction does not go out of control. When an average of more than one neutron collides an atomic explosion occurs. This chain reaction is the basis of the atomic explosion of ^{235}U.

4. Comparison of fission and fusion reactions

Comparison	Fission Reaction	Fusion Reaction
Definition	Heavier nucleus splits into lighter nuclei	Lighter nuclei fuse to give heavier nucleus
Chain reaction	Occurs	Does not occur
Activation energy	Relatively low	Extremely high
Controllability	Easy	Difficult
Use of released energy	Easy	Very hard
Nature of products	Radioactive	Non-radioactive
Solar energy source	No	Yes
Occurance	Artificial	Natural
Nuclear waste	Yes	No
Applications	Nuclear reactor	CERN large hadron collider

Discussion Questions:

- Predict the main factors that determine whether a nuclear reaction proceeds via fusion or fission.
- Nuclear fusion reactions occur within the Sun. Validate the statement.
- Differentiate nuclear fission from fusion reactions in terms of the energy involved.

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